

Packet Tracer - Back Up Configuration Files

Instructor Note: Red font color or gray highlights indicate text that appears in the instructor copy only.

Objectives

Part 1: Establish Connectivity to TFTP Server

Part 2: Transfer the Configuration File from TFTP Server

Part 3: Backup Configuration and IOS to TFTP Server

Background / Scenario

In this activity you will restore a configuration from a backup and then perform a new backup. Due to an equipment failure, a new router has been put in place. Fortunately, backup configuration files have been saved to a Trivial File Transfer Protocol (TFTP) Server. You are required to restore the files from the TFTP Server to get the router back online as quickly as possible.

Instructions

Part 1: Establish Connectivity to the TFTP Server

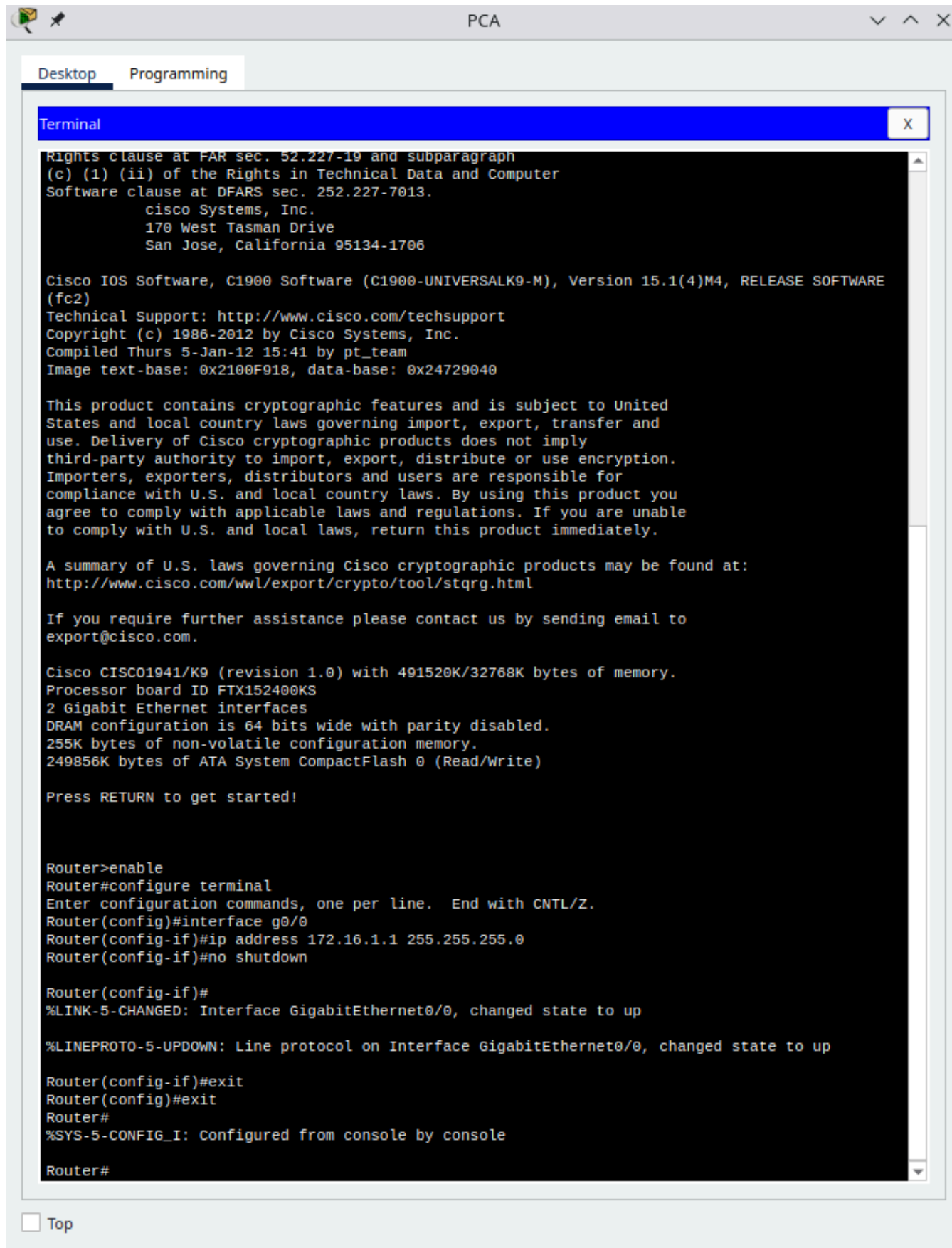
Note: Because this is a new router, the initial configuration will be performed using a console connection to the router.

- Click **PCA**, then the **Desktop** tab, followed by **Terminal** to access the **RTA** command line.
- Configure and activate the **Gigabit Ethernet 0/0** interface. The IP address should match the default gateway for the **TFTP Server**.

```
Router(config)# interface g0/0
```

```
Router(config-if)# ip address 172.16.1.1 255.255.255.0
```

```
Router(config-if)# no shut
```



- c. Test connectivity to **TFTP Server**. Troubleshoot, if necessary.

```
Router#ping 172.16.1.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.1.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
Router#
```

Part 2: Transfer the Configuration File from the TFTP Server

- a. From privileged EXEC mode, issue the following command:

```
Router# copy tftp running-config
Address or name of remote host []? 172.16.1.2
Source filename []? RTA-config
Destination filename [running-config]? <cr>
```

The router should return the following:

```
Accessing tftp://172.16.1.2/RTA-config...
Loading RTA-config from 172.16.1.2: !
[OK - 785 bytes]
785 bytes copied in 0.001 secs
RTA#
%SYS-5-CONFIG_I: Configured from console by console
RTA#
```

- b. Issue the command to display the current configuration.

What changes were made?

The configuration stored on the TFTP Server was loaded into the router and the hostname of the router changed to RTA.

```
Router#
Router#copy tftp running-config
Address or name of remote host []? 172.16.1.2
Source filename []? RTA-config
Destination filename [running-config]?

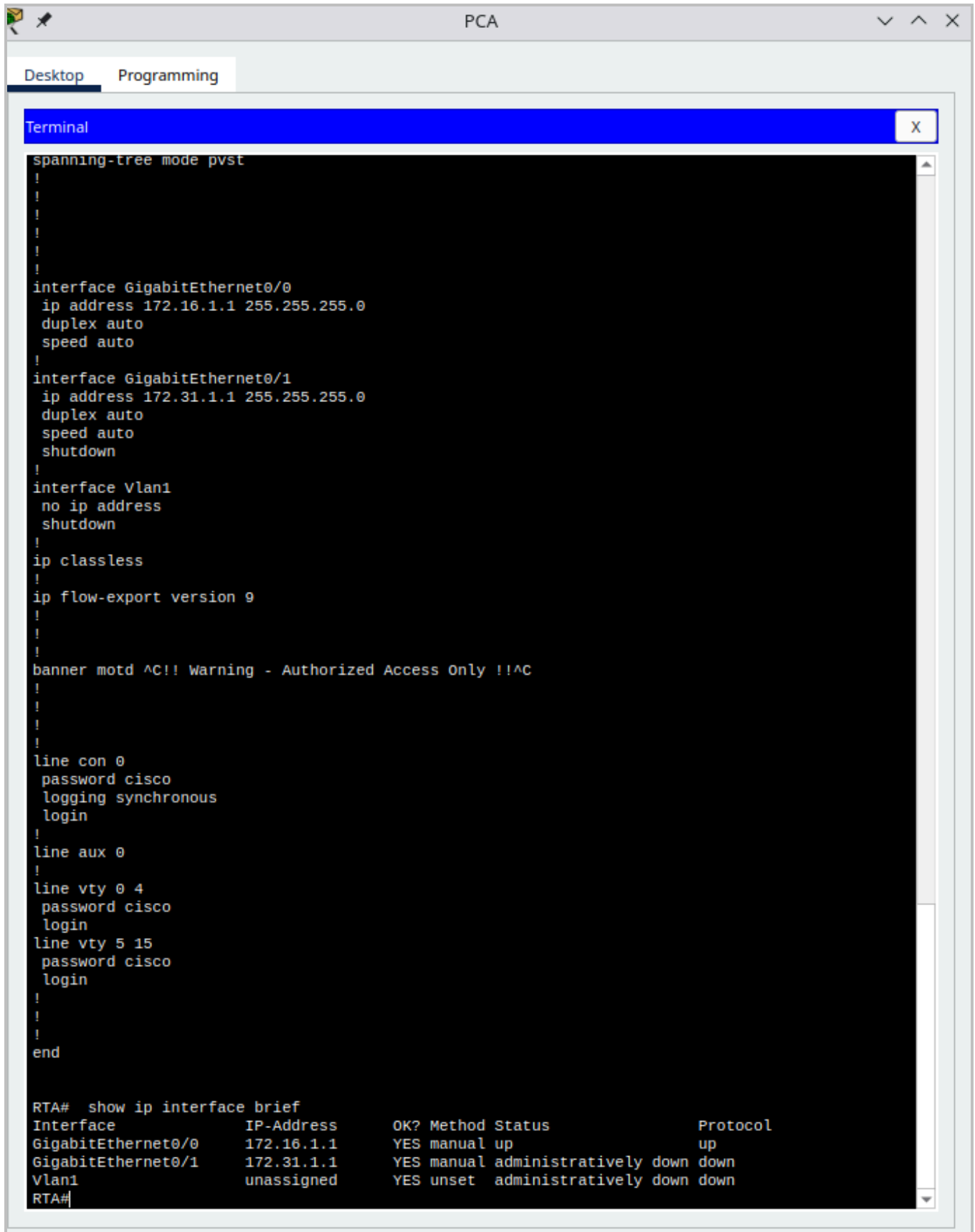
Accessing tftp://172.16.1.2/RTA-config...
Loading RTA-config from 172.16.1.2: !
[OK - 785 bytes]

785 bytes copied in 0 secs
RTA#
%SYS-5-CONFIG_I: Configured from console by console
RTA#
```

- c. Issue the appropriate **show** command to display the interface status.

Are all interfaces active?

No, G0/1 is administratively down.



The screenshot shows a Packet Tracer window titled 'PCA' with a 'Terminal' tab active. The terminal displays a Cisco configuration for a switch named 'RTA'. The configuration includes setting spanning-tree mode to 'pvst', configuring two GigabitEthernet interfaces (0/0 and 0/1) with IP addresses 172.16.1.1 and 172.31.1.1 respectively, and configuring a Vlan1 interface. It also sets the flow-export version to 9, a MOTD banner, and console/VTY passwords. The configuration ends with the 'end' command. Below the configuration, the command 'show ip interface brief' is executed, displaying a table of interface status.

```
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 172.16.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 172.31.1.1 255.255.255.0
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
!
banner motd ^C!! Warning - Authorized Access Only !!^C
!
!
!
!
line con 0
 password cisco
 logging synchronous
 login
!
line aux 0
!
line vty 0 4
 password cisco
 login
line vty 5 15
 password cisco
 login
!
!
!
end

RTA# show ip interface brief
Interface                IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0       172.16.1.1      YES manual up                    up
GigabitEthernet0/1       172.31.1.1      YES manual administratively down down
Vlan1                    unassigned      YES unset  administratively down down
RTA#
```

- d. Correct any issues related to interface problems and test connectivity between PCA and the TFTP server.

```
RTA#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#interface g0/1
RTA(config-if)#no shutdown

RTA(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

RTA(config-if)#exit
RTA(config)#exit
RTA#
%SYS-5-CONFIG_I: Configured from console by console

RTA#
```

Part 3: Back Up Configuration and IOS to TFTP Server

- a. Change the hostname of RTA to RTA-1.
- b. Save the configuration to NVRAM.
- c. Copy the configuration to the **TFTP Server** using the **copy** command:

```
RTA-1# copy running-config tftp:
Address or name of remote host []? 172.16.1.2
Destination filename [RTA-1-config]? <cr>
```

```
RTA#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
RTA(config)#hostname RTA-1
RTA-1(config)#exit
RTA-1#
%SYS-5-CONFIG_I: Configured from console by console

RTA-1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
RTA-1#
```

- d. Issue the command to display the files in flash.
- e. Backup the IOS in flash to the **TFTP Server** using the following command:

```
RTA-1# copy flash tftp:
Source filename []? c1900-universalk9-mz.SPA.151-4.M4.bin
Address or name of remote host []? 172.16.1.2
Destination filename [c1900-universalk9-mz.SPA.151-4.M4.bin]? <cr>
```

What special character repeatedly displays indicating that the IOS file is being copied to the TFTP server successfully?

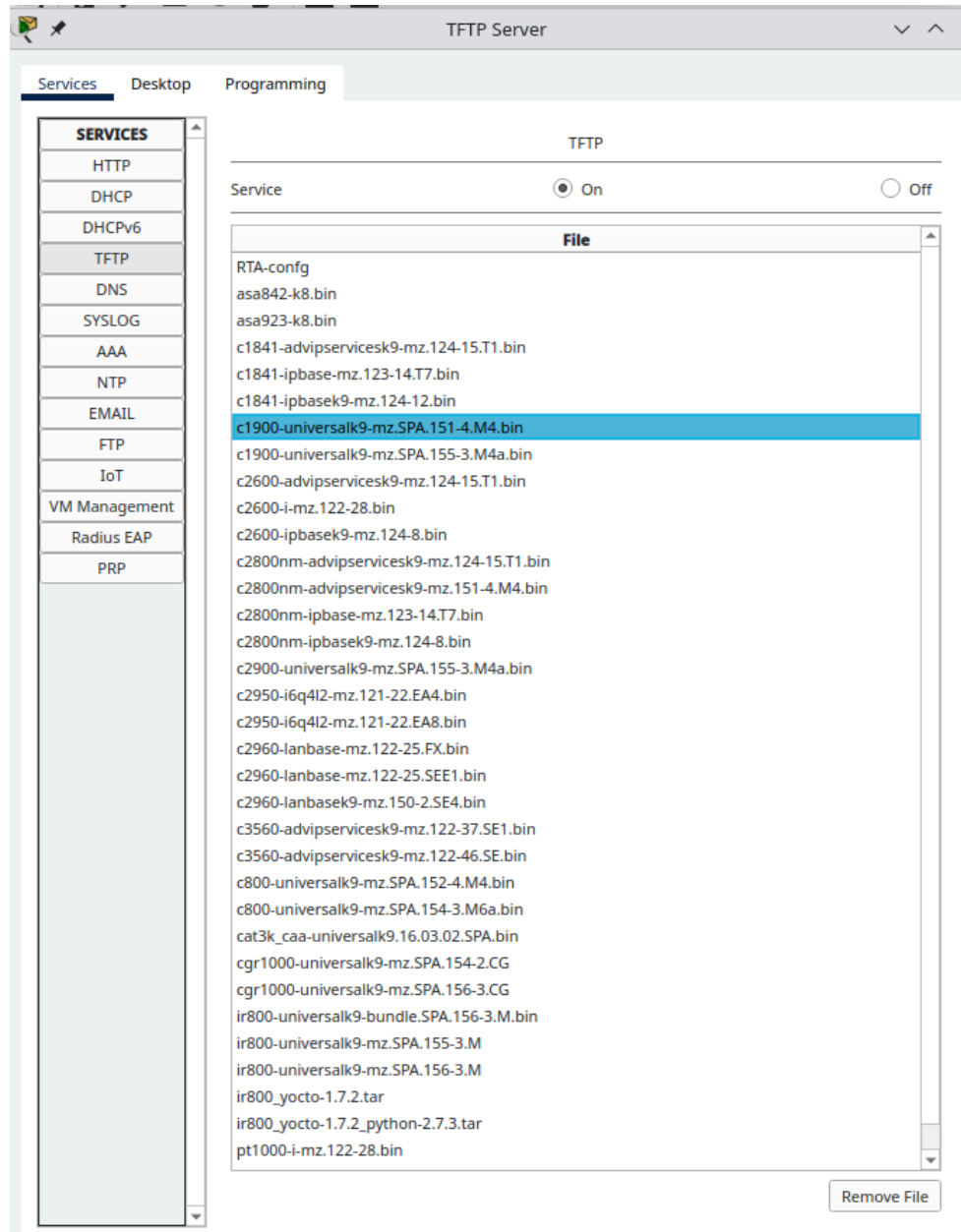
[illegible]

33591768 bytes copied in 0.43 secs (8202318 bytes/sec)

The exclamation point !.

- f. Open the TFTP Server and click the Services tab, select TFTP, and scroll through the list of IOS files. Has the IOS file **c1900-universalk9-mz.SPA.151-4.M4.bin** been copied to the TFTP Server?

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Yes, the c1900-universalk9-mz.SPA.151-4.M4.bin file is listed in the files on the TFP server.